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* Find GCD (750, 900).

→ Formula $\Rightarrow A = B \times Q + R$

where, A = large No.

B = small No.

Q = Quotient

R = remainder

Q	A	B	R
1	900	750	150
5	750	150	0
	150	0	

$$\Rightarrow \text{GCD}(750, 900) = 150$$

* RAS $\Rightarrow$ Asymmetric Public-key Encryption

Ex: $\Rightarrow$ Using RAS algorithm, generate the Public and Private key for $P=17$ and $q=11$.

S: $\Rightarrow$ 1 Two Prime No.

$$P=17, q=11$$

S: $\Rightarrow$ 2 Find the value of n

$$\therefore n = P \times q \\ = 17 \times 11$$

$$\underline{n = 187}$$

S: $\Rightarrow$ 3 $\Rightarrow$ Find $\phi(n)$

$$\therefore \phi(n) = (P-1)(q-1)$$

$$\phi(187) = (17-1)(11-1)$$

$$= 16 \times 10$$

$$\underline{\phi(187) = 160}$$

5:→4 select the Public Exponent e

choose e such that: $1 < e < \phi(n)$

and

$$\text{GCD}(e, \phi(n)) = 1$$

Here, choose:

$$e = 7$$

check:

$$\text{GCD}(7, 160) = 1$$

Therefore,

$$\boxed{e = 7}$$

5:→5 calculate the Private Exponent d

$$d \times e \equiv 1 \pmod{\phi(n)}$$

substitute the value:

$$d \times 7 \equiv 1 \pmod{160}$$

$$d = 23 \text{ because } (7 \times 23 = 161), (161 - 160 = 1)$$

Therefore,

$$7 \times 23 \equiv 1 \pmod{160}$$

$$\boxed{d = 23}$$

5:→6 Generate RSA keys

$$\text{Public key } (e, n) = (7, 187)$$

$$\text{Private key } (d, n) = (23, 187)$$

⇒ Final Answer.

$$\begin{array}{l} P=17 \\ Q=11 \end{array} \left\{ \begin{array}{l} n=187 \\ \phi(n)=160 \end{array} \right\} \left\{ \begin{array}{l} e=7 \\ d=23 \end{array} \right\} \left\{ \begin{array}{l} \text{Public key} = (7, 187) \\ \text{Private key} = (23, 187) \end{array} \right.$$

* Chinese Remainder Theorem

Example $\Rightarrow$ Solve the following system of Congruences using Chinese Remainder theorem.

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

$$\text{Formula} \Rightarrow x = (a_1 m_1 m_1^{-1} + a_2 m_2 m_2^{-1} + a_3 m_3 m_3^{-1}) \pmod{M}$$

$\Rightarrow$ Given,

$$a_1 = 2 \quad \left\{ \begin{array}{l} m_1 = 3 \\ m_2 = 5 \\ m_3 = 7 \end{array} \right.$$

S $\Rightarrow$ 1 Find M ,

$$\therefore M = m_1 \times m_2 \times m_3$$

$$M = 3 \times 5 \times 7$$

$$\underline{M = 105}$$

S $\Rightarrow$ 2 Find M_1, M_2, M_3

$$M_1 = \frac{M}{m_1} = \frac{105}{3} = 35$$

$$M_2 = \frac{M}{m_2} = \frac{105}{5} = 21$$

$$M_3 = \frac{M}{m_3} = \frac{105}{7} = 15$$

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S:→3 → Find m_i^{-1}

$$\begin{cases} m_1 \times m_1^{-1} = 1 \pmod{m_1} \\ 35 \times m_1^{-1} = 1 \pmod{3} \\ 35 \times 2 = 1 \pmod{3} \\ m_1^{-1} = 2 \end{cases} \begin{cases} m_2 \times m_2^{-1} = 1 \pmod{m_2} \\ 21 \times m_2^{-1} = 1 \pmod{5} \\ 21 \times 1 = 1 \pmod{5} \\ m_2^{-1} = 1 \end{cases} \begin{cases} m_3 \times m_3^{-1} = 1 \pmod{m_3} \\ 15 \times m_3^{-1} = 1 \pmod{7} \\ 15 \times 1 = 1 \pmod{7} \\ m_3^{-1} = 1 \end{cases}$$

S:→4) Apply CRT Formula

$$\begin{aligned} \therefore x &= (a_1 m_1 m_1^{-1} + a_2 m_2 m_2^{-1} + a_3 m_3 m_3^{-1}) \pmod{M} \\ x &= (2 \times 35 \times 2 + 3 \times 21 \times 1 + 2 \times 15 \times 1) \pmod{105} \\ x &= (140 + 63 + 30) \pmod{105} \\ x &= 233 \pmod{105} \\ x &= 23 \end{aligned}$$

Final answer $x = 23 \pmod{105}$

* Explain difference between DES vs 3DES vs AES.

	DES	3DES	AES
Full Form	Data Encryption standard	Triple data Encryption standard	Advanced Encryption standard
TYPE	Symmetric - key	block cipher	block cipher
Block size	64 bits	64 bits	128 bits
Key size	56 bits	112 or 168 bits	128, 192 or 256 bits
Encryption method	DES encryption once	DES applied three times	substitution and Permutation based
Speed	Faster than 3DES	Slow	Faster than DES & 3DES
Security	Low/weak	Better than DES	Very High
Structure	Feistel network	Feistel network	Substitution-Permutation network

Current Status	Obsolete	Legacy/deprecated	Widely used
Example	older encryption system	legacy banking/payment system	wifi, VPN, file encryption
Simple Ex. $\Rightarrow$	Data $\rightarrow$ DES $\rightarrow$ Ciphertext	Data $\rightarrow$ DES $\rightarrow$ DES $\rightarrow$ DES $\rightarrow$ Ciphertext	Data $\rightarrow$ AES $\rightarrow$ Ciphertext

* Explain the AES algorithm with neat diagram of its structure.

$\rightarrow$ AES (Advanced Encryption Standard) is a symmetric-key block cipher used for secure encryption and decryption of data. It replaced DES because it provides stronger security and better performance.

- $\rightarrow$ AES works on a 128-bit block of plaintext
- $\rightarrow$ It supports 128-bit, 192-bit and 256-bit keys.
- $\rightarrow$ AES has 10, 12 or 14 rounds, depending on the key size.
- $\rightarrow$ It uses a substitution-permutation network structure.
- $\rightarrow$ The 128-bit data is arranged in a 4×4 matrix of bytes, called the state.
- $\rightarrow$ AES is used in wifi, VPNs, file encryption and secure communication.

$\Rightarrow$ Advantages $\Rightarrow$ High security $\Rightarrow$ AES gives strong protection for data.

$\Rightarrow$ Fast $\Rightarrow$ AES is faster than DES and 3DES.

$\Rightarrow$ Disadvantages $\Rightarrow$ key management $\Rightarrow$ The secret key must be shared safely.

$\Rightarrow$ Implementation Risk $\Rightarrow$ Poor key storage can reduce security.

* Difference between Symmetric and Asymmetric

	Symmetric	Asymmetric
Full form $\rightarrow$	Symmetric Key Cryptography	Asymmetric Key Cryptography
No. of keys $\rightarrow$	uses one key.	uses two key (Public & Private)
key used $\rightarrow$	same key is used for encryption and decryption.	Different keys are used for encryption and decryption.
key sharing $\rightarrow$	secret key must be shared	Public key can be shared openly;
security		Private key is kept secret.
Speed $\rightarrow$	Faster	Slower than symmetric crypto...
security $\rightarrow$	secure secure, but secure key sharing is difficult.	Provides secure key exchange and authentication.
Examples $\rightarrow$	AES, DES	RA, ECC
Example $\rightarrow$	same key $\rightarrow$ Encrypt $\rightarrow$ Decrypt	Public Key $\rightarrow$ Encrypt $\rightarrow$ Private Key $\rightarrow$ Decrypt

* Difference Between Public key and Private key

	Public key	Private key
Meaning $\rightarrow$	Key shared with others	Secret key kept by owner
used in $\rightarrow$	Asymmetric cryptography	Asymmetric cryptography.
visibility $\rightarrow$	Publicly available	must be kept secret
sharing $\rightarrow$	can be freely shared	should never be shared.
main use $\rightarrow$	Encryption	Decryption
Security $\rightarrow$	Does not need to be secret.	Must be protected
used by $\rightarrow$	other can use it.	only owner use it.
Example $\rightarrow$	Public key $\rightarrow$ Encrypt message.	Private key $\rightarrow$ Decrypt message.